

Brownian motion

(MS Project Presentation)

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Table of contents

1. Properties of Brownian motion
2. Construction of Brownian motion
3. The canonical model of a Brownian motion
4. Brownian motion as a martingale
5. Brownian motion as a Markov process
6. The variation of Brownian paths
7. Regularity of Brownian paths
8. Simulation of Brownian motion

Introduction

Let $(\Omega, \mathcal{A}, P)$ be a probability space.

A **d-dimensional Brownian motion** $B = (B_t)_{t \geq 0}$ is a stochastic process (defined on Ω), indexed by $[0, \infty)$ taking values in $\mathbb{R}^d$ such that

1. $B_0(\omega) = 0$ for almost all ω .
2. For all $n \geq 0$, $0 = t_0 < t_1 < t_2 < \dots < t_n < \infty$,
 $B_{t_n} - B_{t_{n-1}}, \dots, B_{t_1} - B_{t_0}$ are independent.
3. $B_t - B_s \sim B_{t+h} - B_{s+h}$ such that $s + h \geq 0$.
4. $B_t - B_s \sim N(0, t - s)^{\otimes d}$, $N(0, t)(dx) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{x^2}{2t}\right) dx$.
5. For all ω , $t \mapsto B_t(\omega)$, $t \in [0, \infty)$ is a continuous function.

We denote a d-dimensional Brownian motion by **BM^d**.

Properties of Brownian motion

Covariance and moments of a BM¹

Let (B_t) be a BM¹.

1. Since, the expectation of B_t is 0 for all t , covariance of B_s and B_t is given by $\mathbb{E}(B_s B_t)$. If $s \leq t$, we observe by definition of Brownian motion that

$$\mathbb{E}(B_s B_t) = \mathbb{E}(B_s (B_t - B_s)) + \mathbb{E}(B_s^2) = s = s \wedge t.$$

2. Since for all $t \geq 0$,

$$\mathbb{E}e^{i\xi B_t} = e^{-\frac{1}{2}\sigma^2\xi^2},$$

it follows that B_t are Gaussian random variables with mean 0 and covariance t .

Corollary

For any $t \geq 0$, exponential moments of all orders of B_t exists, and is given by

$$\mathbb{E}e^{\zeta B_t} = e^{t\zeta^2/2}$$

for all $\zeta \in \mathbb{C}$.

This can be used to calculate the moments of B_t as follows:

Lemma

For all $k = 0, 1, \dots$,

1. $\mathbb{E}(B_t^{2k+1}) = 0$,
2. $\mathbb{E}(B_t^{2k}) = t^k \cdot (2k - 1) \cdot \dots \cdot 3 \cdot 1$.

In particular, we have $\mathbb{E}(B_t) = \mathbb{E}(B_t^3) = 0$, $\mathbb{V}(B_t) = \mathbb{E}(B_t^2) = t$ and $\mathbb{E}(B_t^4) = 3t^2$.

Brownian motion as a Gaussian process

Definition (Gaussian process)

A one-dimensional stochastic process $(X_t)_{t \geq 0}$ is a Gaussian process if for any $n \geq 1$ and any $0 \leq t_1 < t_2 < \dots < t_n$, we have that the vector $\Gamma = (X_{t_1}, \dots, X_{t_n})$ is a Gaussian random vector.

Theorem (Characterization of BM¹)

Let $(B_t)_{t \geq 0}$ be a one-dimensional stochastic process. Then the following are equivalent:

1. $(B_t)_{t \geq 0}$ is a one dimensional Brownian motion.
2. $(B_t)_{t \geq 0}$ is a Gaussian process, such that for all $n \in \mathbb{N}$, $\Gamma = (B_{t_1}, \dots, B_{t_n})$ for $0 \leq t_1 < t_2 < \dots < t_n < \infty$ is a Gaussian random vector with mean 0, covariance matrix $C = (t_i \wedge t_j)_{i,j=1,\dots,n}$, and continuous sample paths.

This characterization is obtained by showing that any such arbitrary random vector Γ (with the properties described above) always follows

$$\mathbb{E} e^{i\langle \xi, \Gamma \rangle} = e^{-\frac{1}{2}\langle \xi, C\xi \rangle}.$$

So, it is obvious to ask for a generalization of this characterization to the d -dimensional case! Such a characterization indeed holds.

Theorem

Let $(X_t)_{t \geq 0}$ be a d -dimensional stochastic process. Then the following are equivalent:

1. $(X_t)_{t \geq 0}$ satisfies axioms 1-4 in the definition of BM^d .
2. For all $n \geq 0$, $0 = t_0 < t_1 < \dots < t_n$ and $\xi_0, \dots, \xi_n \in \mathbb{R}^d$, we have

$$\begin{aligned} & \mathbb{E} \left(\exp \left(i \sum_{j=1}^n \langle \xi_j, X_{t_j} - X_{t_{j-1}} \rangle + i \langle \xi_0, X_{t_0} \rangle \right) \right) \\ &= \exp \left(-\frac{1}{2} \sum_{j=1}^n |\xi_j|^2 (t_j - t_{j-1}) \right). \end{aligned}$$

Corollary (Coordinate processes of BM^d)

Let $B = (B_t)_{t \geq 0}$ be a d -dimensional stochastic process. Then the following are equivalent:

- 1. B is a d -dimensional Brownian motion.*
- 2. The coordinate processes $B = (B^1, \dots, B^d)$ are each 1-dimensional independent Brownian motion.*

Invariance properties of a Brownian motion

Theorem

Let $B = (B_t)_{t \geq 0}$ be a BM^d .

1. **Reflection.** The stochastic process $-B = (-B_t)_{t \geq 0}$ is a BM^d .
2. **Renewal.** Let $a > 0$ be a fixed time. Then the stochastic process $W = (W_t)_{t \geq 0}$ for $W_t = B_{t+a} - B_a$ is a BM^d .
3. **Markov property.** Fix $a > 0$. Let $W = (W_t)_{t \geq 0}$ for $W_t = B_{t+a} - B_a$ be the renewed BM^d . Then $(B_t)_{0 \leq t \leq a}$ and $(W_t)_{t \geq 0}$ are independent.
4. **Time inversion.** Fix $a > 0$. Let $W = (W_t)_{0 \leq t \leq a}$ where $W_t = B_{a-t} - B_a$. Then W is a BM^d .
5. **Scaling.** Fix $c > 0$. For each $t > 0$, we have that $B_{ct} \sim c^{1/2} B_t$. In particular, $(c^{-1/2} B_{ct})_{t \geq 0}$ is again a BM^d .

Construction of Brownian motion

Brownian motion as a random series

Considering $(\Omega, \mathcal{A}, P)$ on which a sequence of iid $N(0, 1)$ random variables $(G_n)_{n \geq 0}$ is supported, and assuming $(\phi_n)_{n \geq 0}$ to be a **complete orthonormal system**, we define

$$W_N(t) := \sum_{n=0}^{N-1} G_n \langle \mathbb{1}_{[0,t]}, \phi_n \rangle_{L^2} = \sum_{n=0}^{N-1} G_n \int_0^t \phi_n(s) ds.$$

Lemma

The limit $W(t) := \lim_{N \rightarrow \infty} W_N(t)$ exists for every $t \in [0, 1]$ in $L^2(P)$ and the process $W(t)$ satisfies the conditions (1-4) in the definition of Brownian motion.

We consider the Haar orthonormal system in the random orthogonal series to show the continuity of sample paths of $W(t)$.

For $n = 0$, and $n = 2^j + k$, $j \geq 0$, $k = 0, 1, \dots, 2^j - 1$, the Haar ($H_n(t)$) and Schauder ($S_n(t)$) systems are defined as follows:

- $H_0(t) = 1$, and,
- $H_{2^j+k} = \begin{cases} +2^{j/2} & \text{if } t \in [\frac{k}{2^j}, \frac{2k+1}{2^{j+1}}) \\ -2^{j/2} & \text{if } t \in [\frac{2k+1}{2^{j+1}}, \frac{k+1}{2^j}) \\ 0 & \text{otherwise.} \end{cases}$
- $S_0(t) = 1$, and,
- $S_{2^j+k} = \int_0^t H_{2^j+k}(s) ds.$

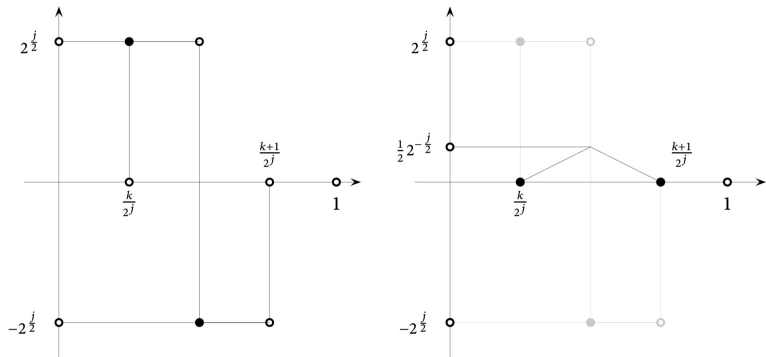


Figure 1: The Haar functions H_{2^j+k} and the Schauder functions S_{2^j+k}

Theorem (Lévy-Ciesielski, 1959)

There is a probability space $(\Omega, \mathcal{A}, P)$ and a sequence of iid standard Gaussian variables $(G_n)_{n \geq 0}$ such that

$$W(t, \omega) = \sum_{n=0}^{\infty} G_n(\omega) < \mathbb{1}_{[0,t]}, H_n >_{L^2},$$

$t \in [0, 1]$ is a Brownian motion.

Corollary

Let $(W^k(t))_{t \in [0,1]}$, $k \geq 0$ be independent real valued Brownian motion defined on the same probability space. Then

$$B(t) := \begin{cases} W^0(t) & \text{if } t \in [0, 1) \\ W^0(1) + W^1(t-1) & \text{if } t \in [1, 2) \\ \sum_{j=0}^{k-1} W^j(1) + W^k(t-k) & \text{if } t \in [k, k+1), k \geq 2. \end{cases}$$

is a Brownian motion indexed by $t \in [0, \infty)$.

Motivation towards constructing the random series: Lévy's argument via interpolation

Consider a BM¹, say $(W_t)_{t \in [0,1]}$. Then, for $t_0 < t < t_1$,

$$G' := W(t) - W(t_0) \quad \text{and} \quad G'' := W(t_1) - W(t_0)$$

are **independent Gaussian random variables** with mean 0 and variances $(t - t_0)$ and $(t_1 - t)$ respectively. Now, consider $\Gamma \sim N(0, 1)$ which is independent to $W(t_0)$ and $W(t_1) - W(t_0)$. Then,

$$\begin{aligned} \Gamma' &= \frac{(t_1 - t_0)W(t_0) + (t - t_0)W(t_1)}{t_1 - t_0} + \sqrt{\frac{(t_1 - t)(t - t_0)}{t_1 - t_0}} \Gamma \\ &= W(t_0) + \left(\frac{t - t_0}{t_1 - t_0} \right) [W(t_1) - W(t_0)] + \sqrt{\frac{(t_1 - t)(t - t_0)}{t_1 - t_0}} \Gamma. \end{aligned}$$

is a Gaussian random variable with mean 0 and variance t .

Since, Γ' has the same distribution as that of $W(t)$, it follows that

$$P(W(t) \in \cdot \mid W(t_0) = x_0, W(t_1) = x_1) = N(m_t, \sigma_t^2)$$

where

$$m_t = \frac{(t_1 - t)x_0 + (t - t_0)x_1}{t_1 - t_0}$$
$$\sigma_t^2 = \frac{(t_1 - t)(t - t_0)}{t_1 - t_0} = \frac{1}{4}(t_1 - t_0), \text{ if } t = \frac{t_1 + t_0}{2}.$$

So, if we consider $t_0 = 0$ and $t_1 = 1$, then, for the **mid-point** $\frac{1}{2}$, we obtain $W(\frac{1}{2}, \omega)$ by adding Γ' to the linear interpolation of $W(0)$ and $W(1)$, that is,

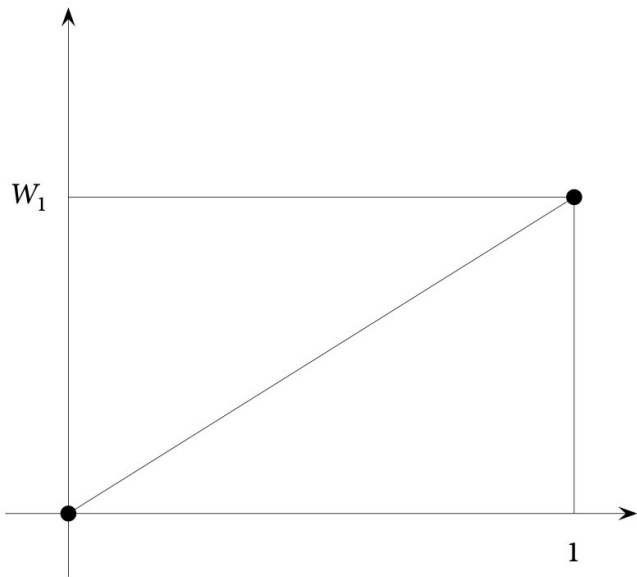
$$W\left(\frac{1}{2}, \omega\right) = \frac{1}{2}[W(0, \omega) + W(1, \omega)] + \Gamma'.$$

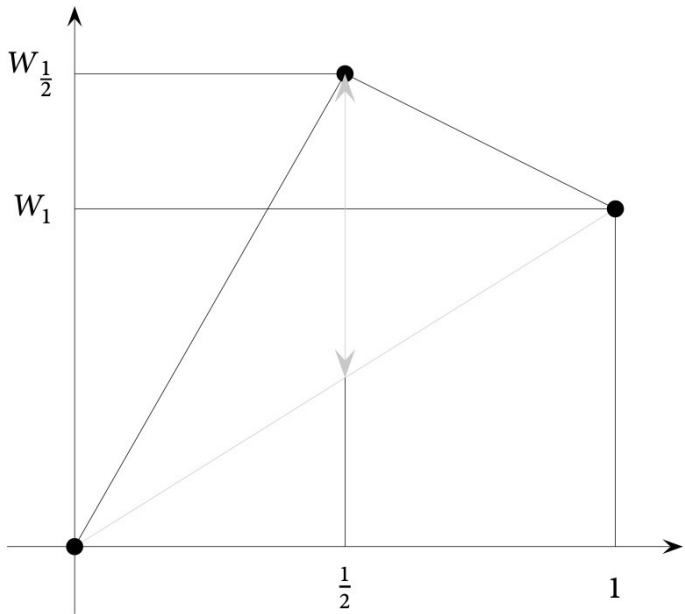
The general algorithm for interpolation

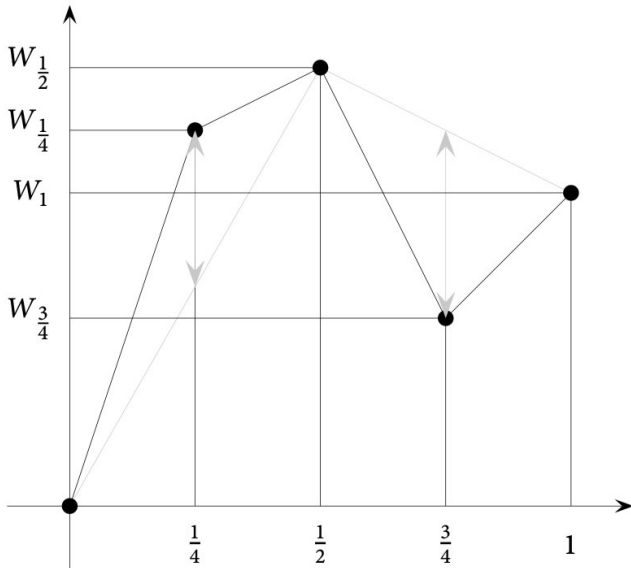
We set $W(0) = 0$ and $W(1) \sim N(0, 1)$. For $n \geq 1$, we assume that the random variables $W(k2^{-n})$, $k = 1, 2, \dots, 2^n - 1$ have already been constructed. Then,

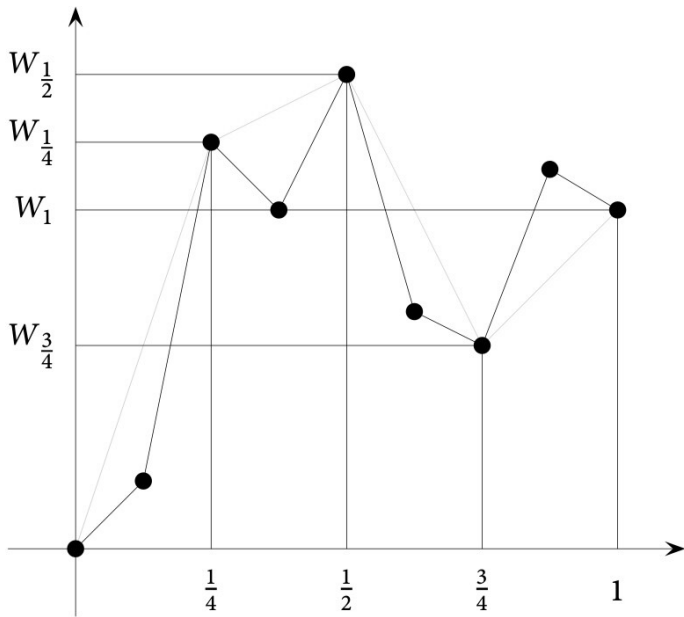
$$W(l2^{-n-1}) := \begin{cases} W(k2^{-n}) & \text{if } l = 2k, \\ \frac{1}{2}[W(k2^{-n}) + W((k+1)2^{-n})] + \Gamma_{2^n+k} & \text{if } l = 2k+1, \end{cases}$$

where Γ_{2^n+k} is an independent (of everything else) Gaussian random variable with mean 0 and variance $2^{-n}/4$.









Define

$W_{2^n}(\omega) :=$ Linear interpolation of $(W(k2^{-n}, \omega), k = 0, 1, \dots, 2^n)$, $n \geq 1$.

Theorem

The series

$$W(t, \omega) := \sum_{n=0}^{\infty} (W_{2^{n+1}}(t, \omega) - W_{2^n}(t, \omega)) + W_1(t, \omega)$$

for $t \in [0, 1]$ converges a.s. uniformly. In particular, $(W(t))_{t \in [0, 1]}$ is a BM^1 .

By corollary 10, we obtain a BM^1 on the entirety of $[0, \infty)$.

The polygonal arc $W_{2^n}(t, \omega)$ can be expressed in terms of Schauder functions as follows:

$$W(t, \omega) = G_0(\omega)S_0(t) + \sum_{n=0}^{\infty} \sum_{k=0}^{2^n-1} G_{2^n+k}(\omega)S_{2^n+k}(t),$$

where,

$$G_0(\omega)S_0(t) = W_1(t, \omega),$$

$$\sum_{k=0}^{2^n-1} G_{2^n+k}(\omega)S_{2^n+k}(t) = W_{2^{n+1}}(t, \omega) - W_{2^n}(t, \omega).$$

and, $S_n(t)$ are the Schauder functions. This shows that Lévy's construction and the Lévy-Ciesielski construction are indeed the same!

The canonical model of a Brownian motion

What is this canonical model about?

Any $\mathbb{R}^d$ -valued stochastic process $(X(t))_{t \in I}$ is a map

$$\begin{aligned} I \times \Omega &\longrightarrow \mathbb{R}^d \\ (t, \omega) &\longmapsto X(t, \omega). \end{aligned}$$

We can also view this map as

$$\begin{aligned} \Omega &\longrightarrow (\mathbb{R}^d)^I = \{w : w : I \rightarrow \mathbb{R}^d\} \\ \omega &\longmapsto \{t \mapsto X(t, \omega)\}. \end{aligned}$$

If we identify Ω with a subset of $(\mathbb{R}^d)^I$, then, we get the so-called **canonical model**. For Brownian motion it is enough to consider the subspace $\mathcal{C}_{(0)}[0, \infty) \subset (\mathbb{R}^d)^{[0, \infty)}$, where,

$$\mathcal{C}_{(0)} := \mathcal{C}_{(0)}[0, \infty) := \{w : I \rightarrow \mathbb{R}^d \text{ is continuous and } w(0) = 0\}.$$

Wiener space and Wiener measure

Let $I = [0, \infty)$, and define

$$\begin{aligned}\pi_t : (\mathbb{R}^d)^I &\rightarrow \mathbb{R}^d \\ w &\mapsto w(t).\end{aligned}$$

We also define the canonical σ -algebra on the infinite product $(\mathbb{R}^d)^I$ as

$$\mathcal{B}^I(\mathbb{R}^d) = \sigma\{\pi_t^{-1}(B) : B \in \mathcal{B}(\mathbb{R}^d), t \in I\} = \sigma\{\pi_t : t \in [0, 1]\}.$$

Since, a Brownian motion has continuous sample paths, we replace $(\mathbb{R}^d)^I$ by $\mathcal{C}_{(0)}$. But, as $\mathcal{C}_{(0)} \not\subseteq \mathcal{B}^I(\mathbb{R}^d)$, we consider the **trace σ -algebra**

$$\mathcal{C}_{(0)} \cap \mathcal{B}^I(\mathbb{R}^d) = \sigma\{\pi_t|_{\mathcal{C}_{(0)}} : t \in [0, 1]\}.$$

Sigma algebra on the space $\mathcal{C}_{(0)}$

We consider the metric of **locally uniform convergence**:

$$\rho(w, v) = \sum_{n=1}^{\infty} \left(1 \wedge \sup_{0 \leq t \leq n} |w(t) - v(t)| \right) 2^{-n}.$$

If $\mathcal{C}_{(0)}$ is equipped with this metric, then the space becomes a complete and separable metric space.

Lemma

If $\mathcal{O}_\rho$ is the topology induced by ρ , then, the Borel σ -algebra $\mathcal{B}(\mathcal{C}_{(0)}) := \sigma(\mathcal{O}_\rho)$ is equal to the previously defined trace σ -algebra, that is,

$$\mathcal{C}_{(0)} \cap \mathcal{B}^l(\mathbb{R}^d) = \mathcal{B}(\mathcal{C}_{(0)}).$$

Measure on the space $\mathcal{C}_{(0)}$: Wiener measure

We consider a $\text{BM}^d (B_t)_{t \geq 0}$ and the map

$$\begin{aligned}\psi : \Omega &\rightarrow \mathcal{C}_{(0)} \\ \omega &\mapsto w := \psi(\omega) = B(-, \omega).\end{aligned}$$

Note that

1. The metric balls $\mathbb{B}^\rho(v, r) \subset \mathcal{C}_{(0)}$ generate $\mathcal{B}(\mathcal{C}_{(0)})$, and
2. $\{\omega : \psi(\omega) \in \mathbb{B}^\rho(v, r)\} = \{\omega : \rho(B(-, \omega), v) < r\}$,

whence, the map

$$\begin{aligned}\omega &\mapsto \rho(B(-, \omega), w) \\ &= \sum_{n=1}^{\infty} \{1 \wedge \sup_{t \in [0, n]} |B(t, \omega) - w(t)|\} 2^{-n} \\ &= \sum_{n=1}^{\infty} \{1 \wedge \sup_{t \in [0, n]} |\pi_t(B(-, \omega)) - \pi_t(w)|\} 2^{-n} \text{ is } \mathcal{A} \text{ measurable.}\end{aligned}$$

Therefore, it follows that ψ is $\mathcal{A}/\mathcal{B}(\mathcal{C}_{(0)})$ measurable, and, the pushforward measure

$$\mu(\Gamma) := P(\psi^{-1}(\Gamma)), \quad \Gamma \in \mathcal{B}(\mathcal{C}_{(0)}),$$

is a measure on $\mathcal{C}_{(0)}$. Now, since the cylinder sets

$$\Gamma = \{w \in \mathcal{C}_{(0)} : \pi_{t_1}(w) \in C_1, \dots, \pi_{t_n}(w) \in C_n\},$$

are $\cap$ -stable generators of $\mathcal{B}(\mathcal{C}_{(0)})$, for all $n \geq 1$, $0 < t_1 < \dots < t_n$, and $C_1, \dots, C_n \in \mathcal{B}(\mathbb{R}^d)$, it follows that the finite dimensional distributions

$$\mu(\Gamma) = \mu(\pi_{t_1} \in C_1, \dots, \pi_{t_n} \in C_n) = P(B(t_1) \in C_1, \dots, B(t_n) \in C_n),$$

uniquely determine the measure μ (this is our Wiener measure!).

This proves the following theorem:

Theorem

On the probability space $(\mathcal{C}_{(0)}, \mathcal{B}(\mathcal{C}_{(0)}), \mu)$, the family of projections $(\pi_t)_{t \geq 0}$ satisfies all the axioms in the definition of BM^d , and thus, $(\pi_t)_{t \geq 0}$ is a Brownian motion.

Definition

The space $(\mathcal{C}_{(0)}, \mathcal{B}(\mathcal{C}_{(0)}), \mu, \pi_t)$ is called the *Wiener space* or *path space*, the measure μ is called the Wiener measure, and $(\pi_t)_{t \geq 0}$ is called the **the canonical (model of the) Brownian motion** process.

Brownian motion as a martingale

Continuous-time martingales

Definition

For $I \subset [0, \infty)$, and $(\mathcal{F}_t)_{t \geq 0}$ a filtration, a martingale $(X_t, \mathcal{F}_t)_{t \in I}$ is a real or complex stochastic process $X_t : \Omega \rightarrow \mathbb{R}^d$ or $X_t : \Omega \rightarrow \mathbb{C}$ satisfying

1. $\mathbb{E}|X_t| < \infty$ for all $t \in I$;
2. X_t is $\mathcal{F}_t$ measurable for all $t \in I$;
3. $\mathbb{E}(X_t | \mathcal{F}_s) = X_s$ for all $s, t \in I, s \leq t$.

Definition (Sub-or supermartingale)

If X_t is real valued, and in property (3) of the definition of a martingale if we have ' $\geq$ ' or ' $\leq$ ', then $(X_t, \mathcal{F}_t)_{t \in I}$ is called sub-or supermartingale respectively.

Definition (Adapted stochastic process)

A stochastic process $(X_t)_{t \in I}$ is said to be adapted to the filtration $(\mathcal{F}_t)_{t \in I}$ if for each $t \in I$, X_t is $\mathcal{F}_t$ -measurable.

Some 'Brownian' martingales

The **canonical filtration** for a Brownian motion $(B_t)_{t \in I}$ is given by

$$\mathcal{F}_t^B := \sigma(\mathcal{F}_s : s \leq t).$$

Since, from the Markov property of the Brownian motion B_t , we have

$$B_t - B_s \perp\!\!\!\perp \mathcal{F}_s^B \text{ for all } 0 \leq s \leq t, \quad (1)$$

it follows that a Brownian motion is a martingale with respect to its natural filtration $\mathcal{F}_t^B$. The first two properties of the definition are clear; for property (3),

$$\mathbb{E}(B_t | \mathcal{F}_s) = \mathbb{E}(B_t - B_s | \mathcal{F}_s) + \mathbb{E}(B_s | \mathcal{F}_s) = \mathbb{E}(B_t - B_s) + B_s = B_s.$$

For a d -dimensional Brownian motion $(B_t)_{t \geq 0}$, a filtration $(\mathcal{F}_t)_{t \geq 0}$ is called **admissible**, if

1. $\mathcal{F}_t^B \subset \mathcal{F}_t$ for all $t \geq 0$,
2. $B_t - B_s \perp \mathcal{F}_s$ for all $0 \leq s \leq t$.

Also, if $\mathcal{F}_0$ contains all the $\mathbf{P}$ null sets, then, $(\mathcal{F}_t)_{t \geq 0}$ is called *admissible complete filtration*.

Example

Let $(B_t)_{t \geq 0}$ be a d -dimensional Brownian motion and $(\mathcal{F}_t)_{t \geq 0}$ be an admissible filtration.

1. $(B_t)_{t \geq 0}$ is a martingale with respect to $\mathcal{F}_t$.
2. $M_t := |B_t|^2$ is a positive submartingale with respect to $\mathcal{F}_t$.
3. $M_t := |B_t|^2 - d \cdot t$ is a martingale with respect to $\mathcal{F}_t$.

Characterization of a BM^d through conditional characteristic functions

Lemma

Let $(X_t)_{t \geq 0}$ be a $\mathbb{R}^d$ -valued continuous stochastic process with $X_0 = 0$ and which is adapted to the filtration $(\mathcal{F}_t)_{t \geq 0}$. Then $(X_t, \mathcal{F}_t)_{t \geq 0}$ is a Brownian motion, if and only if, for all $0 \leq s \leq t$ and $\xi \in \mathbb{R}^d$

$$\mathbb{E}[e^{i\langle \xi, X_t - X_s \rangle} | \mathcal{F}_s] = e^{-\frac{1}{2}|\xi|^2(t-s)}.$$

Stopping and sampling

A random time $\tau : \Omega \rightarrow [0, \infty)$ is called a **stopping time** if it exhibits the property

$$\{\omega \in \Omega : \tau(\omega) \leq t\} \in \mathcal{F}_t \text{ for all } t \geq 0,$$

that is, in other words, the event $\{\tau \leq t\}$, is completely determined by the random variables $X_s, s \leq t$.

Some examples of stopping times are entry and hitting times of a process $(X_t)_{t \geq 0}$ into a set $A \in \mathcal{B}(\mathbb{R}^d)$, defined as:

1. **first entry time** into A : $\tau_A^\circ(\omega) := \inf\{t \geq 0 : X_t(\omega) \in A\}$,
2. **first hitting time** of A : $\tau_A(\omega) := \inf\{t > 0 : X_t(\omega) \in A\}$,

where if the sets are empty, then we define $\inf \emptyset := \infty$. Clearly, $\tau_A^\circ \leq \tau_A$.

Entry time and hitting time for open & closed sets

Since, the canonical filtration of a stochastic process is relatively small, we consider a **slightly larger filtration**

$$\mathcal{F}_{t+}^X := \bigcap_{u>t} \mathcal{F}_u^X = \bigcap_{n \geq 1} \mathcal{F}_{t+\frac{1}{n}}^X.$$

Lemma

Let $(X_t)_{t \geq 0}$ be a d -dimensional stochastic process with right continuous sample paths and $U \subset \mathbb{R}^d$ an open set. The first hitting time τ_U satisfies

$$\{\omega : \tau_U(\omega) < t\} \in \mathcal{F}_t^X \text{ and } \{\omega : \tau_U(\omega) \leq t\} \in \mathcal{F}_{t+}^X.$$

Since, τ_U° and τ_U coincide, τ_U° is also a $\mathcal{F}_{t+}^X$ stopping time.

Lemma

Let $(X_t)_{t \geq 0}$ be a d -dimensional stochastic process with continuous sample paths and $F \subset \mathbb{R}^d$ be a closed set. Then τ_F° is a $\mathcal{F}_t^X$ stopping time whereas τ_F is a $\mathcal{F}_{t+}^X$ stopping time.

Point recurrence property of a Brownian motion

Let $(B_t)_{t \geq 0}$ be a BM¹, then the first entry time $\tau_b := \tau_{\{b\}}^\circ$ into the closed set $\{b\}$ is known as the **first passage time**.

Now, we have

$$\sup_{t \geq 0} B_t \geq \sup_{n \geq 1} B_n = \sup_{n \geq 1} (\xi_1 + \cdots + \xi_n) =: S,$$

where $\xi_j := B_j - B_{j-1}$ are iid $N(0, 1)$ random variables, and since $S = \infty$ a.s., applying the same argument to $\inf_{t \geq 0} B_t$, we obtain

$$P \left(\sup_{t \geq 0} B_t = +\infty, \inf_{t \geq 0} B_t = -\infty \right) = 1, \quad (2)$$

and hence, $P(\tau_b < \infty) = 1$. Equation 2 implies that a BM oscillates between $+\infty$ and $-\infty$, whence, the random set $\{t \geq 0 : B_t(\omega) = b\}$ is a.s. unbounded, for all $b \in \mathbb{R}$, that is, a BM is point recurrent.

Optional stopping time theorem

Theorem

Let $(X_t, \mathcal{F}_t)_{t \geq 0}$ be a martingale with continuous sample paths and let $\sigma \leq \tau$ be $\mathcal{F}_t$ stopping times. Then, we have that for all $k \geq 0$, $X_{\sigma \wedge k}$, $X_{\tau \wedge k} \in L^1(P)$, and

$$\mathbb{E}(X_{\tau \wedge k} | \mathcal{F}_{\sigma \wedge k}) = X_{\sigma \wedge k}.$$

Moreover, if $\sigma \leq \tau$ are bounded $\mathcal{F}_t$ stopping times, that is, τ is an a.s. bounded stopping time, $X_\sigma, X_\tau \in L^1(P)$, and $\lim_{k \rightarrow \infty} \mathbb{E}[|X_k(\omega)| \chi_{\{\omega: \tau(\omega) > k\}}] = 0$, then

$$\mathbb{E}(X_\tau | \mathcal{F}_\sigma) = X_\sigma.$$

Wald's Identity

Theorem

Let $(B_t, \mathcal{F}_t)_{t \geq 0}$ be a BM^1 and assume that τ is an $\mathcal{F}_t$ stopping time. If $\mathbb{E} \tau < \infty$, then $B_\tau \in L^2(P)$ and we have

$$\mathbb{E} B_\tau = 0, \quad \text{and} \quad \mathbb{E} B_\tau^2 = \mathbb{E} \tau.$$

Corollary

Let $(B_t)_{t \geq 0}$ be a BM^1 and τ_1 be the first passage time of the level 1. Then $\mathbb{E} \tau_1 = \infty$.

Proof.

We have already seen that the first passage time $\tau_1 < \infty$ a.s. Then, $B_{\tau_1} = 1$ which implies the expectation $\mathbb{E} B_{\tau_1} = 1$ and the second moment $\mathbb{E} B_{\tau_1}^2 = 1$. But by Wald's identity, this can only happen if $\mathbb{E} \tau_1 = \infty$. $\square$

Brownian motion as a Markov process

Markov property of a BM

Let us define

$$W_t := B_{t+s} - B_s.$$

Then, considering the canonical filtrations $\mathcal{F}_s^B$ and $\mathcal{F}_\infty^W := \sigma(B_{t+s} - B_s : t \geq 0)$ of B_t and W_t respectively, we have

$$\mathcal{F}_s^B \amalg \mathcal{F}_\infty^W.$$

Theorem

Let $(B_t)_{t \geq 0}$ be a BM^d and $(\mathcal{F}_t)_{t \geq 0}$ be an admissible filtration. Then for every $s > 0$, the process W_t , $t \geq 0$ is also a BM^d and $(W_t)_{t \geq 0}$ is independent of $\mathcal{F}_s$, that is, $\mathcal{F}_\infty^W \amalg \mathcal{F}_s$.

The above theorem implies that we can always **split up our Brownian motion into two independent components**

$$B(t+s) = B(t+s) - B(s) + B(s) = W(t) + y|_{y=B_s}.$$

Notations for the renewed BM

1. We set

$$P^x(B_{t_1} \in A_1, \dots, B_{t_n} \in A_n) := P(B_{t_1} + x \in A_1, \dots, B_{t_n} + x \in A_n),$$

where $0 \leq t_1 < \dots < t_n$ and $A_1, \dots, A_n \in \mathcal{B}(\mathbb{R}^d)$. This means that $P^x(B_s \in A)$ denotes the probability that the Brownian motion starts at the point x and traverses s units of time into the set A .

2. Analogously, we define

$$\mathbb{E}^x B_t = \mathbb{E}(B_t + x).$$

Henceforth, P^x is a **well-defined measure** on $(\Omega, \mathcal{F}_\infty)$ and the map $x \mapsto \mathbb{E} u(B_t + x)$, for all $u \in \mathcal{B}_b(\mathbb{R}^d)$ is measurable!

Result from conditional expectations

Lemma

If $\mathcal{X}$ and $\mathcal{Y}$ are two independent sub- σ algebras of $\mathcal{A}$, and X and Y are $\mathcal{X}/\mathcal{B}(\mathbb{R}^d)$ and $\mathcal{Y}/\mathcal{B}(\mathbb{R}^d)$ measurable random variables, then for all bounded Borel measurable functions $\Phi : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$, we have

$$\mathbb{E} [\Phi(X, Y)|\mathcal{X}] = \mathbb{E} [\Phi(x, Y)|_{x=X}] = \mathbb{E} [\Phi(X, Y)|X]. \quad (3)$$

If $\Psi : \mathbb{R}^d \times \Omega \rightarrow \mathbb{R}$ is bounded and $(\mathcal{B}(\mathbb{R}^d) \otimes \mathcal{Y})/\mathcal{B}(\mathbb{R})$ measurable, then

$$\mathbb{E} [\Psi(X(.), .)|\mathcal{X}] = \mathbb{E} [\Psi(x, .)|_{x=X}] = \mathbb{E} [\Psi(X(.), .)|X]. \quad (4)$$

Theorem (Markov property)

Let $(B_t)_{t \geq 0}$ be a BM^d with admissible filtration $(\mathcal{F}_t)_{t \geq 0}$. Then

$$\mathbb{E} [u(B_{t+s}) | \mathcal{F}_s] = \mathbb{E} [u(B_t + x)] |_{x=B_s} = \mathbb{E}^{B_s} [u(B_t)] \quad (5)$$

holds for all bounded measurable functions $u : \mathbb{R}^d \rightarrow \mathbb{R}$, and

$$\mathbb{E} [v(B_{\cdot+s}) | \mathcal{F}_s] = \mathbb{E} [v(B_{\cdot} + x)] |_{x=B_s} = \mathbb{E}^{B_s} [v(B_{\cdot})] \quad (6)$$

holds for all bounded $\mathcal{B}(\mathcal{C})/\mathcal{B}(\mathbb{R})$ measurable functions $v : \mathcal{C}[0, \infty) \rightarrow \mathbb{R}$ which may depend upon the whole Brownian path.

Proof.

Equation 5 follows from 3 by taking $X = B_s$, $Y = B_{t+s} - B_s$, $\mathcal{X} = \mathcal{F}_s$ and the function $\Phi(X, Y) = u(x + y)$.

Equation 6 follows from 4 if we take $\mathcal{Y} = \mathcal{F}_{\infty}^W$, where $W_t = B_{t+s} - B_s$ and $\mathcal{X}, X$ are same as above. □

The strong Markov property of a BM

Now, we will show that equation 5 remains valid even if we replace s by $\sigma(\omega)$.

In general, if $(X_t)_{t \geq 0}$ is a stochastic process and σ a stopping time, then, we set

$$X_\sigma(\omega) := \begin{cases} X_{\sigma(\omega)}(\omega) & \text{if } \sigma(\omega) < \infty, \\ \lim_{t \rightarrow \infty} X_t(\omega) & \text{if } \sigma(\omega) = \infty, \\ 0 & \text{in all other cases.} \end{cases}$$

Theorem (Strong Markov property)

Let $(B_t)_{t \geq 0}$ be a BM^d with admissible filtration $(\mathcal{F}_t)_{t \geq 0}$ and $\sigma < \infty$ be a.s. finite stopping time. Then $(W_t)_{t \geq 0}$ defined by $W_t := B_{t+\sigma} - B_\sigma$ is again a BM^d , which is independent of $\mathcal{F}_{\sigma+}$.

Strong Markov property corresponding to stopping times

Theorem

Let $(B_t)_{t \geq 0}$ be a BM^d with admissible filtration $(\mathcal{F}_t)_{t \geq 0}$, and let σ be a stopping time. Then

$$\mathbb{E} [u(B_{t+\sigma}) | \mathcal{F}_{\sigma+}] (\omega) = \mathbb{E} [u(B_t + x)] |_{x=B_\sigma(\omega)} = \mathbb{E}^{B_\sigma(\omega)} [u(B_t)] \quad (7)$$

holds for all bounded measurable functions $u : \mathbb{R}^d \rightarrow \mathbb{R}$, and P almost all $\omega \in \{\sigma < \infty\}$. Also,

$$\mathbb{E} [v(B_{\cdot+\sigma}) | \mathcal{F}_{\sigma+}] = \mathbb{E} [v(B_{\cdot} + x)] |_{x=B_\sigma} = \mathbb{E}^{B_\sigma} [v(B_{\cdot})] \quad (8)$$

holds for all bounded $\mathcal{B}(\mathcal{C})/\mathcal{B}(\mathbb{R})$ measurable functions $v : \mathcal{C}[0, \infty) \rightarrow \mathbb{R}$ (which may depend upon the whole Brownian path) and P almost all $\omega \in \{\sigma < \infty\}$.

Let $\tau(\omega) = \tau(B_{\cdot}(\omega))$ be a **stopping time which is expressed as a functional of a Brownian path**, for example, a first hitting time. Let σ be a further stopping time such that $\sigma \leq \tau$ a.s., and we define a stopping time $\tau' = \tau(B_{\cdot+\sigma})$. Now, if we set $W_{\cdot} := B_{\cdot+\sigma} - B_{\sigma}$, the stopping time τ' can now be expressed as $\tau' = \tau(W_{\cdot} + B_{\sigma})$ and note that for any $u \in \mathcal{B}_b(\mathbb{R}^d)$, the functional $u(B_{\tau})$ and $u(W_{\tau'} + B_{\sigma})$ have the same distribution.

Corollary

Let $(B_t)_{t \geq 0}$ be a BM^d , and $\tau = \tau(B_{\cdot})$ a first hitting time and σ a stopping time such that $\sigma \leq \tau$ a.s. Set $\tau' = \tau(B_{\cdot+\sigma})$ and $W_{\cdot} := B_{\cdot+\sigma} - B_{\sigma}$. Then

$$\mathbb{E} [u(B_{\tau}) | \mathcal{F}_{\sigma+}] = \mathbb{E} [u(W_{\tau'} + x)] |_{x=B_{\sigma}(\omega)} = \mathbb{E}^{B_{\sigma}(\omega)} [u(W_{\tau'})]$$

holds for all $u \in \mathcal{B}_b(\mathbb{R}^d)$ and $\mathbf{P}$ almost all $\omega \in \{\tau < \infty\}$.

Desiré André's reflection principle

For $\text{BM}^1 (B_t)_{t \geq 0}$, let $\tau_b = \inf\{t \geq 0 : B_t = b\}$. Now, since,

$$W_s + b = B_{\tau_b+s} - B_{\tau_b} + b = B_{\tau_b+s}$$

is again a BM, $P(B_t \geq b) = P(B_t < b)$.

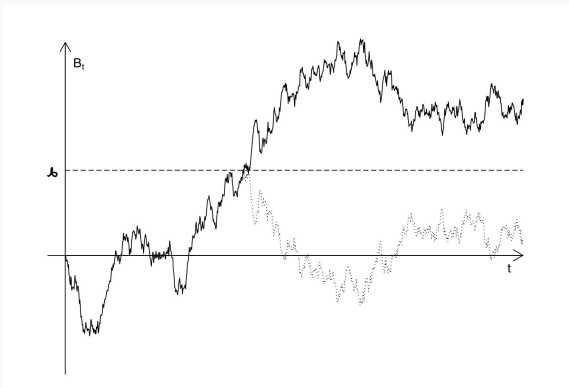


Figure 2: Reflection along the axis $B_{\tau_b} = b$.

We can write

$$P(\tau_b \leq t, B_t < b) = P(\{\tau_b \leq t\} \cap \{B_{\tau_b+(t-\tau_b)} - B_{\tau_b} < 0\}) = \frac{1}{2}P(\tau_b \leq t),$$

since, $\{\tau_b \leq t\} \in \mathcal{F}_{\tau_b}^B$ and $\{B_{\tau_b+(t-\tau_b)} - B_{\tau_b} < 0\} \in \mathcal{F}_{\infty}^W \amalg \mathcal{F}_{\tau_b}^B$. Hence,

$$\begin{aligned} P(\tau_b \leq t) &= P(\tau_b \leq t, B_t \geq b) + P(\tau_b \leq t, B_t < b) \\ &= P(\tau_b \leq t, B_t \geq b) + \frac{1}{2}P(\tau_b \leq t) \\ &= P(B_t \geq b) + \frac{1}{2}P(\tau_b \leq t). \end{aligned}$$

This shows that

$$P(\tau_b \leq t) = 2P(B_t \geq b).$$

Henceforth, defining $M_t := \sup_{s \leq t} B_s$, and observing that $\{\tau_b \leq t\} = \{M_t \geq b\}$, we have,

$$P(M_t \geq b) = P(\tau_b \leq t) = 2P(B_t \geq b) \text{ for all } b > 0.$$

The variation of Brownian paths

p -variation sums

Consider $\Pi = \{0 = t_0 < t_1 < \cdots < t_n = t\}$ a finite partition of the interval $[0, t]$, then, $|\Pi| = \max_{1 \leq j \leq n} (t_j - t_{j-1})$ is called the **fineness/mesh** of the partition. For $p > 0$, the **p -variation sum** is defined to be

$$S_p^\Pi(f; t) := \sum_{t_{j-1}, t_j \in \Pi} |f(t_j) - f(t_{j-1})|^p = \sum_{j=1}^n |f(t_j) - f(t_{j-1})|^p.$$

The **strong/total p -variation** is defined to be

$$\text{VAR}_p(f; t) = \sup\{S_p^\Pi(f; t) : \Pi \text{ is a finite partition of } [0, t]\}.$$

Now, if $(\Pi_n)_{n \geq 1}$ is a sequence of finite partitions of $[0, t]$ such that $\lim_{n \rightarrow \infty} |\Pi_n| = 0$, then

$$\text{var}_p^\Pi(f; t) = \lim_{|\Pi_n| \rightarrow 0} S_p^{\Pi_n}(f; t).$$

Quadratic variation: L^2 -convergence

Theorem

Let $(B_t)_{t \geq 0}$ be a BM¹ and $(\Pi_n)_{n \geq 0}$ be a sequence of finite partitions of $[0, t]$ such that $\lim_{n \rightarrow \infty} |\Pi_n| = 0$. Then, the mean square limit exists:

$$\text{var}_2(B; t) = L^2(P) - \lim_{n \rightarrow \infty} S_2^{\Pi_n}(B; t) = t.$$

Corollary

Almost all Brownian paths are of infinite total variation. In fact, for all $p < 2$, $\text{Var}_p(B; t) = \infty$ a.s.

Indeed for some $\delta > 0$, considering $p = 2 - \delta$, and a sequence $(\Pi_n)_{n \geq 1}$ such that $|\Pi_n| \rightarrow 0$, the result follows by the following inequality:

$$\sum_{j=1}^n (B(t_j) - B(t_{j-1}))^2 \leq \max_{t_j, t_{j-1} \in \Pi_n} |B(t_j) - B(t_{j-1})|^\delta \text{Var}_{2-\delta}(B; t).$$

(Local) Hölder continuity

Definition ((locally) Hölder continuous)

A function $f : [0, \infty) \rightarrow \mathbb{R}$ is said to be (locally) Hölder continuous of order $\alpha > 0$ if for all compact intervals $[a, b] \subset [0, \infty)$, there exists a constant $c = c(f, \alpha, [a, b])$ such that

$$|f(t) - f(s)| \leq c|t - s|^\alpha \text{ for all } s, t \in [a, b].$$

Corollary

Almost all Brownian paths are nowhere locally Hölder continuous for all $\alpha > \frac{1}{2}$.

Quadratic variation: a.s. convergence

Theorem (Lévy, 1948)

Let $(B_t)_{t \geq 0}$ be a BM^1 . For any sequence of finite partitions $(\Pi_n)_{n \geq 0}$ of $[0, t]$ which satisfies $\sum_{n=1}^{\infty} |\Pi_n| < \infty$, we have

$$\text{var}_2(B; t) = \lim_{n \rightarrow \infty} S_2^{\Pi_n}(B; t) = t \quad \text{almost surely.}$$

Note that $n|\Pi_n| \leq \sum_{j=1}^n |\Pi_j| \leq \sum_{j=1}^{\infty} |\Pi_j| < \infty$, hence, $|\Pi_n| = O(1/n)$.

Theorem (Dudley, 1973)

Let $(B_t)_{t \geq 0}$ be a BM^1 and $|\Pi_n|$ be any sequence of partitions of $[0, t]$ which satisfies $|\Pi_n| = o(1/\log n)$. Then,

$$\text{var}_2(B; t) = \lim_{n \rightarrow \infty} S_2^{\Pi_n}(B; t) = t \quad \text{almost surely.}$$

Almost sure divergence of the variance sum

Theorem (Lévy, 1948)

Let $(B_t)_{t \geq 0}$ be a BM^1 . For almost all ω , there exists a sequence of nested partitions $(\Pi_n(\omega))_{n \geq 1}$ such that $|\Pi_n(\omega)| \rightarrow 0$ as $n \rightarrow \infty$, and

$$\text{var}_2(B(., \omega); 1) = \lim_{n \rightarrow \infty} S_2^{\Pi_n(\omega)}(B(., \omega); 1) = \infty \quad \text{almost surely.}$$

Corollary

Let $(B_t)_{t \geq 0}$ be a BM^1 . Then, $\text{VAR}_2(B(., \omega); 1) = \infty$ for almost all $\omega \in \Omega$.

Lévy's martingale characterization of Brownian motion

Corollary

Let $(X_t, \mathcal{F}_t)_{t \geq 0}$ be a continuous real valued martingale such that $(X_t^2 - t, \mathcal{F}_t)_{t \geq 0}$ is also a martingale. Then for all partitions $(\Pi_n)_{n \geq 1}$ of $[0, t]$ satisfying $\lim_{n \rightarrow \infty} |\Pi_n| = 0$,

$$\text{var}_2(X; t) = L^2(P) - \lim_{n \rightarrow \infty} S_2^{\Pi_n}(X; t) = t.$$

Theorem (Lévy, 1948)

Let $(X_t, \mathcal{F}_t)_{t \geq 0}$ be a real valued martingale with continuous sample paths such that $(X_t^2 - t, \mathcal{F}_t)_{t \geq 0}$ is also a martingale. Then, $(X_t)_{t \geq 0}$ is a Brownian motion.

Regularity of Brownian paths

Theorem (Payley, Weiner and Zygmund, 1931)

Let $(B_t)_{t \geq 0}$ be a BM^1 . Then, the path $t \mapsto B_t(\omega)$ is for almost all $\omega \in \Omega$ nowhere differentiable in t .

Proof.

The proof is based on showing the complement of the set

$$A_n := \{\omega \in \Omega : B(\cdot; \omega) \text{ is nowhere differentiable in } [0, n]\}$$

is a null set. □

Hölder continuity of Brownian paths

Theorem (Kolmogorov 1934; Slutsky 1937; Chentsov 1956)

Let $(\xi(x))_{x \in \mathbb{R}^n}$ be a stochastic process on $(\Omega, \mathcal{A}, P)$ with values in $\mathbb{R}^d$ and index set $\mathbb{R}^n$. If

$$\mathbb{E}[|\xi(x) - \xi(y)|^\alpha] \leq c|x - y|^{n+\beta}, \quad \text{for all } x, y \in \mathbb{R}^n, \quad (9)$$

holds for some constant $c > 0$, and $\alpha, \beta > 0$, then there exists a modification $(\xi'(x))_{x \in \mathbb{R}^n}$ with exclusively continuous sample paths. Moreover,

$$\mathbb{E} \left[\left(\sup_{\substack{0 < |x-y| < 1 \\ x, y \in [-T, T]^n}} \frac{|\xi'(x) - \xi'(y)|}{|x - y|^\gamma} \right)^\alpha \right] < \infty, \quad (10)$$

for all $T > 0$, and $0 \leq \gamma < \beta/\alpha$. In particular, $x \mapsto \xi'(x)$ is a.s. Hölder continuous for order γ .

Hölder continuity of Brownian paths for order $< 1/2$

Corollary

A d -dimensional Brownian motion $(B_t)_{t \geq 0}$ is Hölder continuous up to order $\gamma < 1/2$.

Proof. Since, any R^d valued Brownian motion is Hölder continuous iff its components are Hölder continuous, it is enough to consider $d = 1$. So, we have

$$\mathbb{E} [|B(t) - B(s)|^{2k}] = c_n |t - s|^k.$$

Now, we apply the previous theorem to obtain the result. (Note that since Brownian paths are already continuous, we need not consider any such modification of B_t). So, according to the previous theorem, we have $\alpha = 2k$, and $\beta = k - 1$. Then

$$\frac{\beta}{\alpha} = \frac{k - 1}{2k} \longrightarrow \frac{1}{2} \text{ as } k \rightarrow \infty.$$

Now, following the proof of the above theorem, for each $\gamma < \frac{k-1}{2k}$, we have an exceptional set $N_k = \Omega \setminus \Omega_k$, where $P(N_k) = 0$. Since, being Hölder continuous of order γ implies being Hölder continuous of order γ' , for all $\gamma' < \gamma$, the assertion for all $\gamma < \beta/\alpha$ follows by considering the null set $N = \cup_{k \geq 1} N_k$. □

Lévy's modulus of continuity

A modulus of continuity is a **positive, continuous and strictly increasing function** $\sigma : [0, \infty) \rightarrow [0, \infty)$ such that $\sigma(0) = 0$.

We consider a non-random function $f : [0, \infty) \rightarrow \mathbb{R}^d$ and the following estimate:

$$|f(t) - f(s)| \leq \sigma(|s - t|), \text{ for all } s, t \geq 0. \quad (11)$$

The following theorem shows that $\sigma(t) := \sqrt{2t \log \frac{1}{t}}$ is the modulus of continuity for a BM^1 .

Theorem (Lévy 1937)

Let $(B_t)_{t \geq 0}$ be a BM^1 . Then,

$$P \left(\limsup_{h \rightarrow 0} \frac{\sup_{0 \leq t \leq 1-h} |B(t+h) - B(t)|}{\sqrt{2h \log \frac{1}{h}}} = 1 \right) = 1.$$

Hölder continuity of Brownian paths for order $1/2$

Corollary

Almost all sample paths of one-dimensional Brownian motion are nowhere Hölder continuous of order $\alpha = \frac{1}{2}$.

Proof.

On $[0, 1)$, it follows by the following observation:

$$\frac{\sup_{0 \leq t \leq 1-h} |B(t+h) - B(t)|}{\sqrt{h}} = \frac{\sup_{0 \leq t \leq 1-h} |B(t+h) - B(t)|}{\sqrt{2h \log \frac{1}{h}}} \times \sqrt{2 \log \frac{1}{h}}.$$

This generalizes to the interval $[0, \infty)$ just by considering the BM $W(t) := B(t+h) - B(t)$ and applying the same proof as above. $\square$

Simulation of Brownian motion

Independent increments

Algorithm 1 Independent increments

Require: $n \geq 0$, $0 = t_0 < t_1 < \dots < t_n$

$b_0 \leftarrow 0$

$j = 1$

while $j \leq n$ **do**

$y \sim N(0, t_j - t_{j-1})$

$b_{t_j} \leftarrow b_{t_{j-1}} + y$

$j \leftarrow j + 1$

end while

At the end of the algorithm, we get $(b_0, b_{t_1}, \dots, b_{t_n}) \sim (B_0, B_{t_1}, \dots, B_{t_n})$, as required.

Interpolation via Levy's argument

Algorithm 2 Interpolation via Levy's argument

Require: $s_0 < s < s_1$, $(b_{s_0}, b_{s_1}) \sim (B_{s_0}, B_{s_1})$

$$b_s \sim N\left(\frac{(s_1-s)b_{s_0} + (s_0-s)b_{s_1}}{s_1-s_0}, \frac{(s-s_0)(s_1-s)}{s_1-s_0}\right)$$

At the end of the algorithm, we get

$$(b_{s_0}, b_s, b_{s_1}) \sim (B_{s_0}, B_s, B_{s_1})$$

under the condition that $B_{s_0} = b_{s_0}$ and $B_{s_1} = b_{s_1}$.

Algorithm 3 Interpolation via Levy's argument

Require: $J \geq 1$

$b_0 \leftarrow 0$

$j \leftarrow 0, l \leftarrow 0$

while $j \leq J - 1$ **do**

while $l \leq 2^j - 1$ **do**

$y \sim N(0, 1)$

$b_{\frac{2l+1}{2^{j+1}}} \leftarrow \frac{1}{2} \left(b_{\frac{l}{2^j}} + b_{\frac{l+1}{2^j}} \right) + 2^{-(\frac{j}{2}+1)} y$

$l \leftarrow l + 1$

end while

$j \leftarrow j + 1$

end while

At the end of the algorithm, we get that

$$(b_0, b_{1/2^J}, b_{2/2^J}, \dots, b_1) \sim (B_0, B_{1/2^J}, B_{2/2^J}, \dots, B_1).$$

[1]



René L Schilling and Lothar Partzsch.

Brownian motion: an introduction to stochastic processes.

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